Homework

Assigning Instrumental Variables
Case of Bulgarian Communist Era Wages (1986)

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Class Model

Specification

Dependent Variable: LOG(WAGE)
Method: Two-Stage Least Squares
Date: 03/19/26
Sample: 1 10500 IF WAGE $>$ 0
Included Observations: 10331
Instrument Specification: C, GRADES, FL, READ, RELIG

Class Model

Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.372735	0.156603	2.380133	0.0173
EDU	-0.033558	0.023950	-1.401175	0.1612
INNOV	0.011635	0.022960	0.506721	0.6124
MAJOR	0.530028	0.020033	26.45774	0.0000
MALE	0.238713	0.019011	12.55664	0.0000
BKP	0.404941	0.197271	2.052710	0.0401
AGE	0.099950	0.003029	32.99719	0.0000
AGE^2	-0.001028	2.69E-05	-38.18487	0.0000

R-squared	0.307434	Mean dependent variable	2.667893
Adjusted R-squared	0.306964	S.D. dependent var	0.829038
S.E. of regression	0.690164	Sum squared residuals	4917.112
F-statistic	696.8929	Durbin-Watson statistic	1.959189
Prob(F-statistic)	0.000000	Second-Stage SSR	4776.209
J-statistic	147.8605	Instrument rank	10
Prob(J-statistic)	0.000000		

Class Model

Exogeneity Test

Equation: CLASS_EQ Workfile: COMMUNISM BCP::Untitled\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Endogeneity Test
Equation: CLASS_EQ
Endogenous variables to treat as exogenous: EDU BKP
Specification: LOG(WAGE) C EDU INNOV MAJOR MALE BKP AGE AGE^2
Instrument specification: C GRADES FL INNOV MAJOR MALE READ
RELIG AGE AGE^2
Null hypothesis: EDU BKP are exogenous

	Value	df	Probability
Difference in J-stats	156.6681	2	0.0000

J-statistic summary:

	Value
Restricted J-statistic	312.4926
Unrestricted J-statistic	155.8245

Class Model

Stock Yogo

Equation: CLASS_EQ Workfile: COMMUNISM BCP::Untitled\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Weak Instrument Diagnostics
Equation: CLASS_EQ

Cragg-Donald F-stat:	28.82042

Stock-Yogo TSLS critical values (relative bias):

5%	11.04
10%	7.56
20%	5.57
30%	4.73

Stock-Yogo critical values (size):

10%	16.87
15%	9.93
20%	7.54
25%	6.28

Moment selection criteria:

SIC-based:	129.3746
HQIC-based:	138.9205
Relevant MSC:	-9.857866

Best Model

Specification

Dependent Variable: LOG(WAGE)
Method: Two-Stage Least Squares
Date: 03/19/26
Sample: 1 10500 IF WAGE > 0
Included Observations: 10331
Instrument Specification: C, GRADES, GRADES^2 FL, FL^2 INNOV, INNOV^2, RELIG

Best Model

Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.750072	0.143508	-5.226673	0.0000
EDU	0.667379	0.068620	9.725742	0.0000
EDU^2	-0.103346	0.009839	-10.50400	0.0000
AGE	0.098050	0.002982	32.88007	0.0000
AGE^2	-0.000998	2.77E-05	-35.99323	0.0000
BKP	0.448531	0.170358	2.632866	0.0085
MAJOR	0.725095	0.026382	27.48472	0.0000
MALE	0.183775	0.020093	9.146081	0.0000
READ	0.148789	0.029493	5.044949	0.0000

R-squared	0.236575	Mean dependent var	2.667893
Adjusted R-squared	0.235983	S.D. dependent var	0.829038
S.E. of regression	0.724646	Sum squared resid	5420.201
F-statistic	574.4696	Durbin-Watson stat	1.953576
Prob(F-statistic)	0.000000	Second-Stage SSR	4686.561
J-statistic	38.03247	Instrument rank	13
Prob(J-statistic)	0.000000		

Best Model

Exogeneity Test

Equation: BEST_EQ Workfile: COMMUNISM BCP::Untitled\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Endogeneity Test
Equation: BEST_EQ
Endogenous variables to treat as exogenous: EDU EDU^2 BKP
Specification: LOG(WAGE) C EDU EDU^2 AGE AGE^2 BKP MAJOR MALE READ
Instrument specification: C GRADES GRADES^2 FL FL^2 INNOV INNOV^2 AGE AGE^2 RELIG MAJOR MALE READ
Null hypothesis: EDU EDU^2 BKP are exogenous

	Value	df	Probability
Difference in J-stats	307.4267	3	0.0000

J-statistic summary:

	Value
Restricted J-statistic	351.6197
Unrestricted J-statistic	44.19301

Best Model

Stock Yogo

Equation: BEST_EQ Workfile: COMMUNISM BCP::Untitled\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Weak Instrument Diagnostics
Equation: BEST_EQ

Cragg-Donald F-stat:	24.59088

Stock-Yogo TSLS critical values (relative bias):

5%	13.95
10%	8.50
20%	5.56
30%	4.44

Stock-Yogo size critical values not available for models with more than 2 endogenous variables.

Moment selection criteria:

SIC-based:	1.060853
HQIC-based:	20.15267
Relevant MSC:	-9.065218

Ending remarks

I am delighted for your attention.

Q & A?